EVALUATING INTEGRALS

Completing the square to reveal a standard form

Many integrals that look new are standard forms in disguise; completing the square reveals this one.

BEFORE

Quadratic that does not factor

$$\int \frac{1}{x^2 + 4x + 8}\,dx$$

Discriminant 16 − 32 = −16 < 0: no real roots to split the fraction with

COMPLETE THE SQUARE

$$x^2 + 4x + 8 = (x + 2)^2 + 4$$

FINISH THE INTEGRAL

AFTER

Standard form ∫ du/(u² + a²)

$$\int \frac{1}{(x + 2)^2 + 4}\,dx$$

Matches with u = x + 2 and a = 2

WHAT CHANGED

The variable part went into $u^2 = (x + 2)^2$, and the constant $4 = 2^2$ became a^2.

WHAT STAYED

The integrand: $x^2 + 4x + 8$ and $(x + 2)^2 + 4$ are the same quadratic.